

Specific Factor Model

特定要素模型

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The **Specific Factors Model** (also known as the **Ricardo-Viner model**, developed by Jacob Viner's (1930s) extensions of Ricardo; formally developed by Ronald Jones (1971) and Peter Neary (1978) bridges the gap between **short-run reality** and **long-run trade theory**. While the Heckscher-Ohlin model assumes all factors are perfectly mobile **across sectors**, the Specific Factors model recognizes that in the short-to-medium run, capital, land, and specialized equipment are often **stuck** in particular industries. Only labor (or broadly defined "mobile factors") can reallocate freely.

The Specific-Factor Model explains why owners of sector-specific capital (e.g., factory owners, farmers, miners) often have strong opinions about trade policy — they gain or lose heavily depending on whether their sector **expands** or **contracts**. This shift fundamentally changes trade's **distributional politics**: winners and losers are determined by sector, not by factor type.

Core Assumptions

Assumption	Description
Two goods (X and Y)	e.g., Manufacturing and Agriculture
Three factors	Labor (L) + Specific Capital in X (K_x) + Specific Capital in Y (K_y)
Labor is mobile between sectors	Workers can move from one industry to another
Capital is specific (immobile)	Capital is stuck in its sector in the short run
Constant Returns to Scale	In each sector
Perfect Competition	In goods and labor markets
Diminishing Marginal Returns	To labor in each sector (due to fixed specific capital)
Full employment of all factors	Labor and both specific capitals

Most Important Feature: Labor is the **mobile factor**, while capital is **sector-specific**.

Main Mechanism

Goods Price → Factor Price Relationship

- An increase in the **price of a good** benefits the **specific factor** used in that sector **strongly**.
- It benefits **labor** (mobile factor) **moderately**, while it **hurts** the specific factor in the **other sector**.

Detailed Effects:

Change	Effect on Wage (w)	Effect on Return to $K_x(r_x)$	Effect on Return to $K_y(r_y)$
Price of X ↑ (e.g. P_x ↑)	Increases	Strong Increase	Decreases
Price of Y ↑ (e.g. P_y ↑)	Increases	Decreases	Strong Increase

Intuition:

- When P_x rises $\rightarrow$ **Demand** for labor in sector X increases $\rightarrow$ wages rise. Sector X expands, pulling labor from sector Y .
- **Owners** of K_x (specific to X) gain a lot (higher revenue + lower real wage cost relative to output price), while **Owners** of K_y lose because their sector contracts.

Real Income Effect:

- Specific factor owners are heavily exposed to price changes in their sector.
- Workers (mobile) have **ambiguous** real income effects depending on their consumption basket.

Endowment Change → Factor Price and Output

A. Increase in Labor Endowment ($\uparrow L$)

- Output of **both goods** increases. Wage (**w**) **falls** (due to diminishing returns).
- Return to both specific capitals (r_x and r_y) **increase**.

B. Increase in Specific Capital (e.g., $\uparrow K_x$)

- Output of X **increases significantly**, while output of Y **decreases** (labor moves to X).
- Wage (**w**) **increases**.
- Return to K_x (r_x) **falls** (diminishing returns), while Return to K_y (r_y) **increases**.

This is similar to the **Rybczynski effect** but modified due to specificity.

Intuition

How Labor Allocation Works:

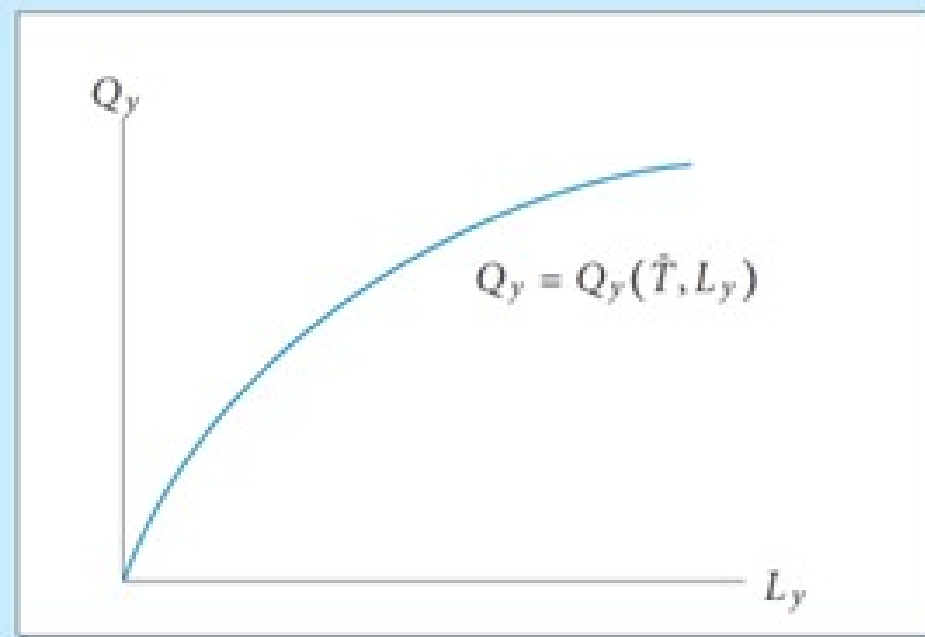
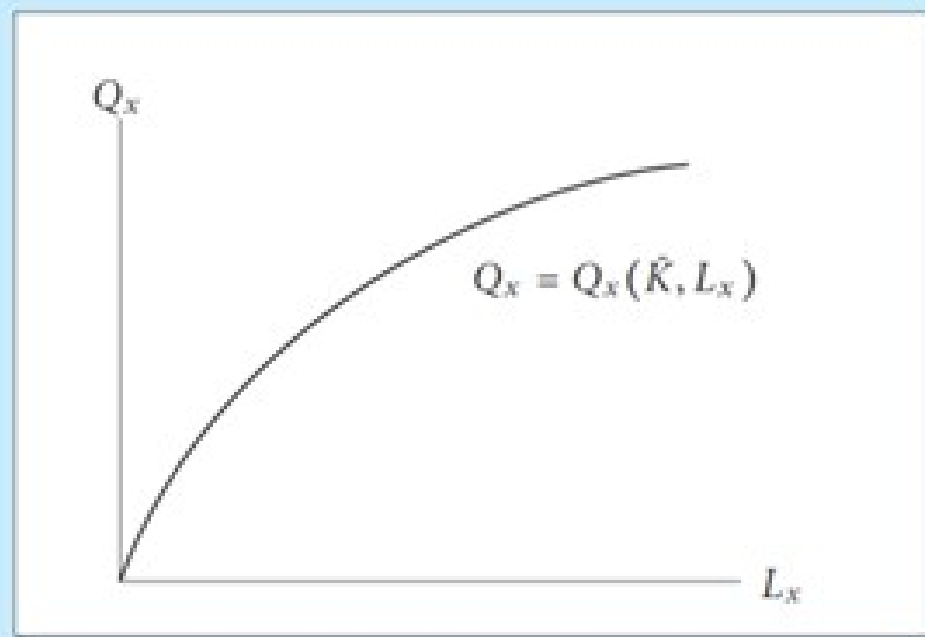
1. Firms maximize profit: hire labor until **Value of Marginal Product of Labor (VMPL)** equals wage:

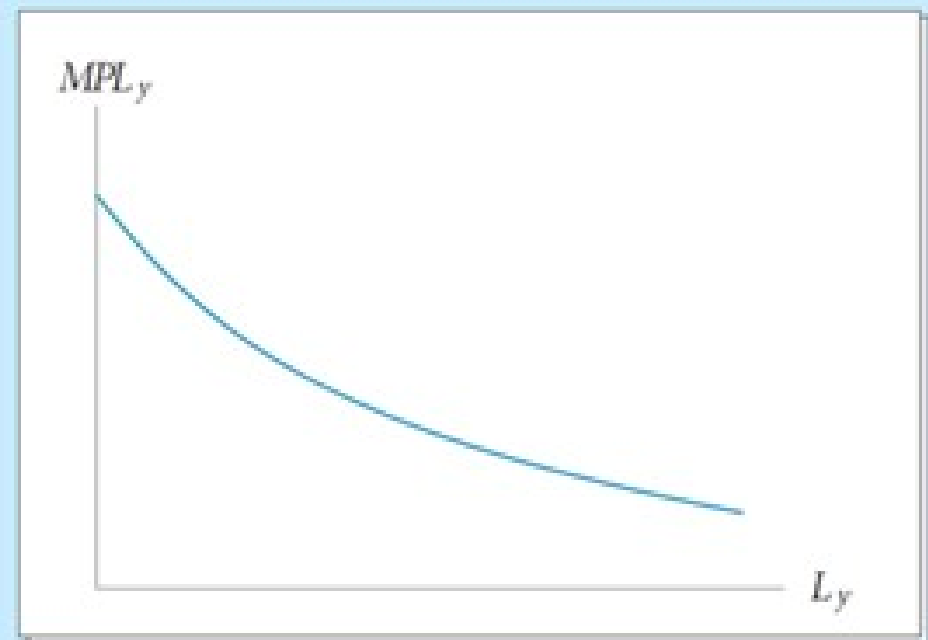
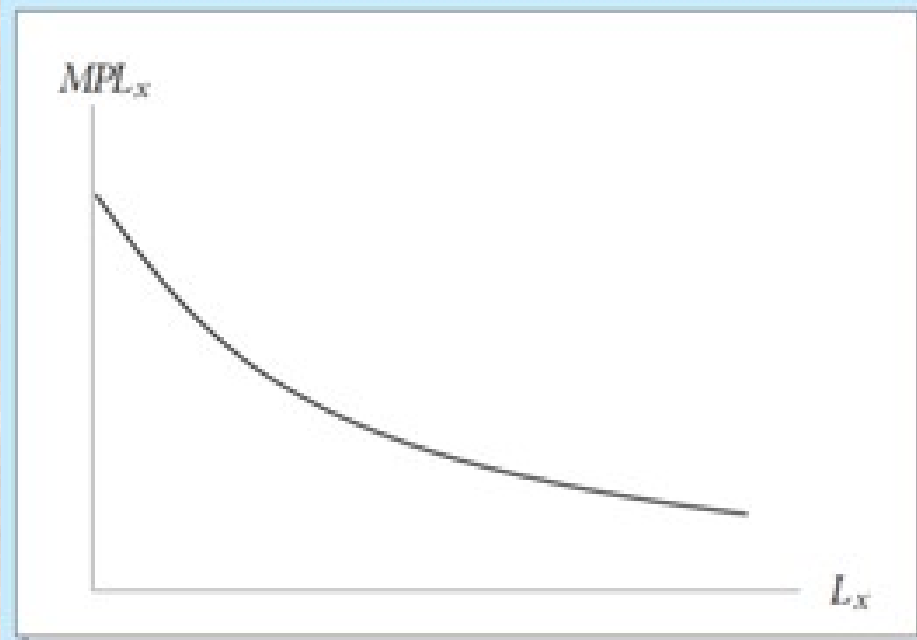
$$VMPL_x = P_X \cdot MPL_X(L_X) = w$$

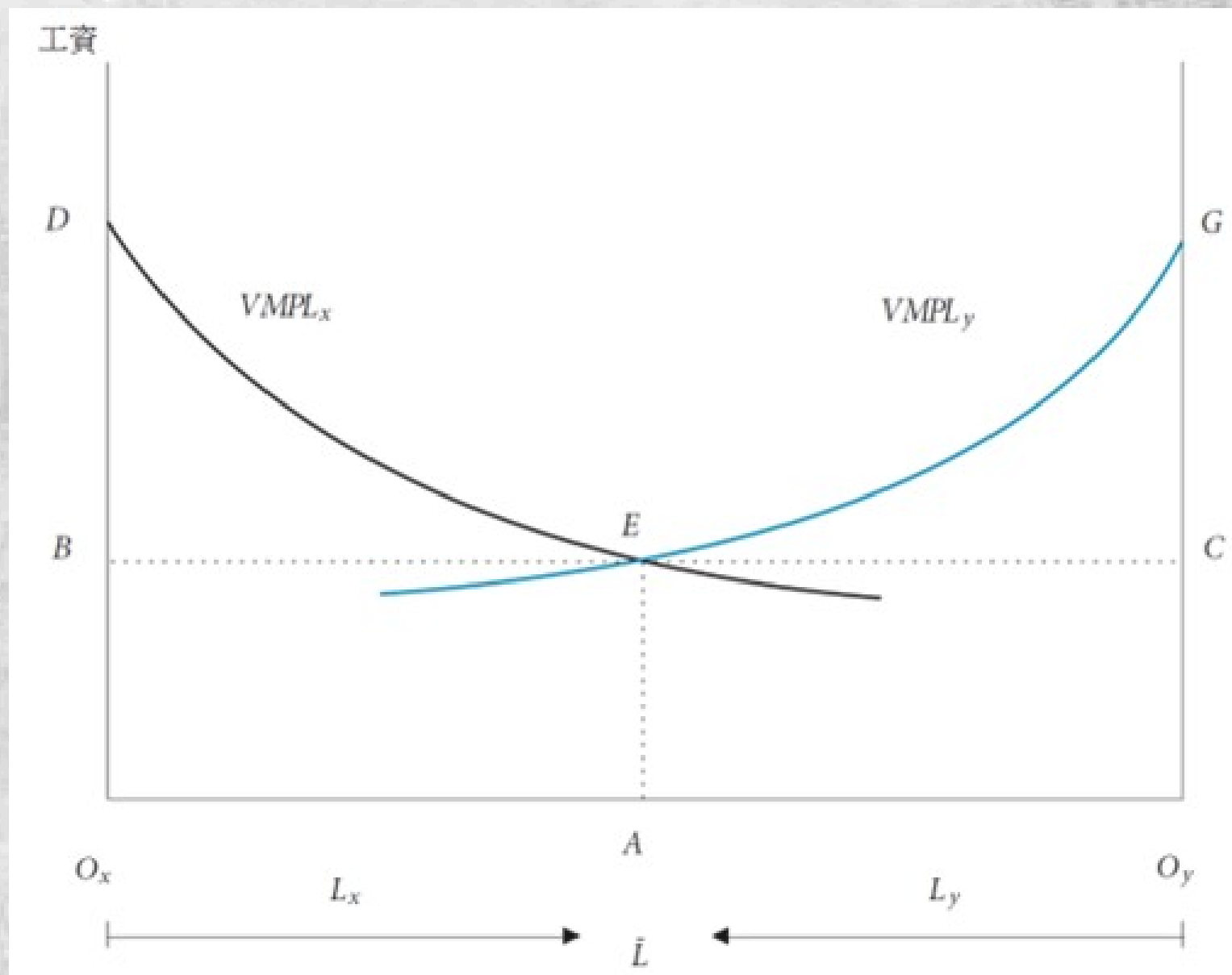
$$VMPL_Y = P_Y \cdot MPL_Y(L_Y) = w$$

MPL **diminishes** as more labor is added to a fixed specific factor $\Rightarrow VMPL$ curves slope downward.

Equilibrium: Labor splits so $VMPL_X = VMPL_Y = w$. The **intersection** of the two $VMPL$ curves determines L_X^*, L_Y^*, w^* .

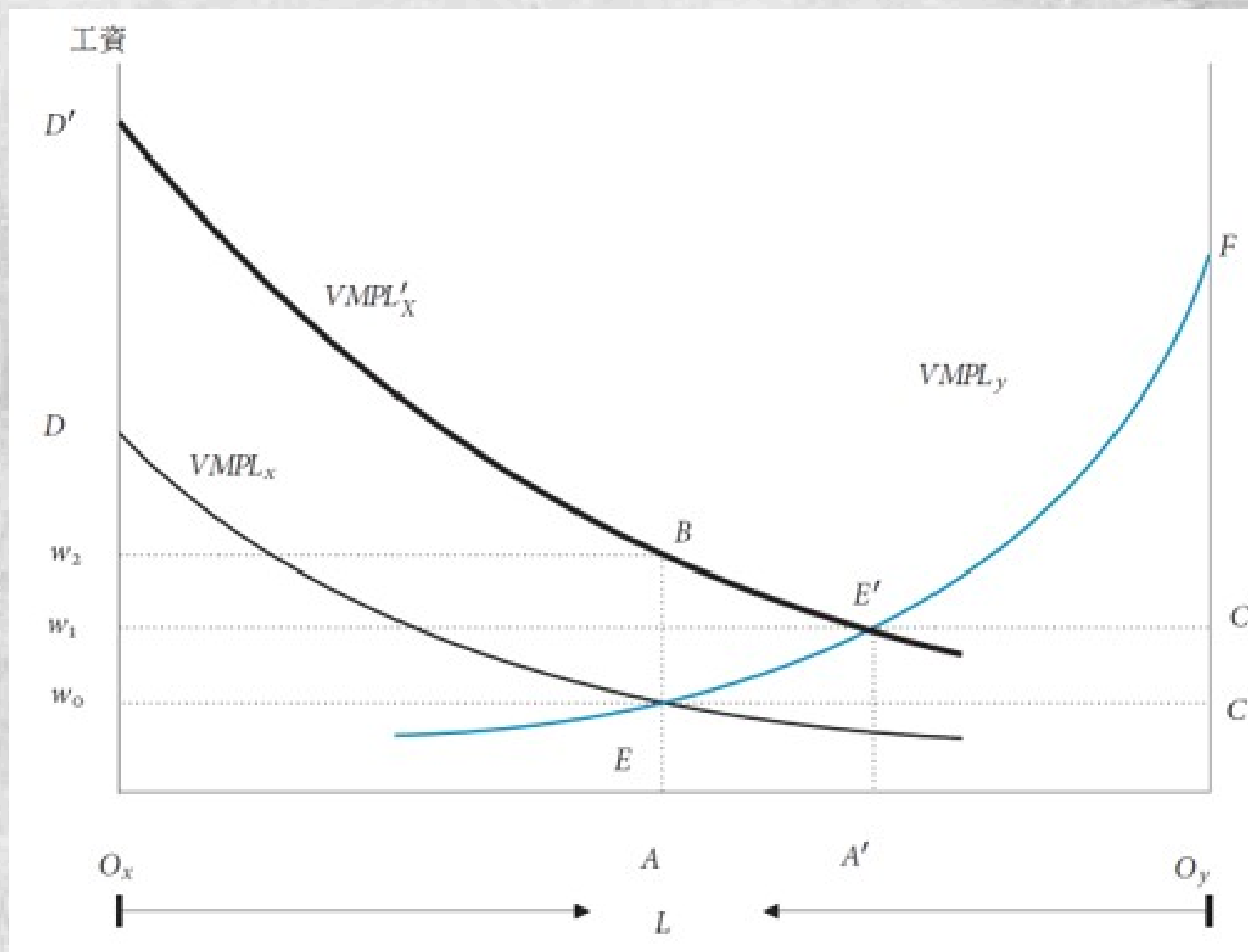






Response to a Price Change (e.g., $P_X \uparrow$ due to trade/tariff):

- $VMPL_X$ shifts upward.
- Labor moves from Y to X until $VMPL_X = VMPL_Y$ again.
- **Wage rises**, but **less than proportionally** to P_X .
- **Sector X expands, Sector Y contracts.**
- Distributional outcomes follow directly from factor ownership patterns.



Summary Comparison with H-O Model

Feature	Specific-Factor Model	Heckscher-Ohlin Model
Time Horizon	Short Run	Long Run
Factor Mobility	Labor mobile, Capital specific	Both factors fully mobile
Effect of Price Change	Strong effect on specific factors	Benefits abundant factor
Income Distribution	Clear winners and losers	Based on factor abundance
Pattern of Trade	Based on specific factor abundance	Based on overall K/L ratio

Key Theorems & Policy Implications

Result	Implication
Sectoral Distribution	Trade creates winners/losers by industry, not by factor class
Ambiguous Labor Effect	Workers gain purchasing power for imported goods, lose for exported goods
Compensation Principle	Aggregate national income rises; gainers could theoretically compensate losers
Tariff Protection	Shields sector-specific assets from foreign competition; explains industry lobbying
Short vs Long Run	Short run: specific factors dominate politics. Long run: capital becomes mobile $\Rightarrow\Rightarrow$ H-O/S-S logic applies

Political Economy Insight:

The model explains why trade politics often align with **sectoral coalitions** (e.g., `steelworkers + steel owners` vs. `auto workers + auto owners`) rather than class-based factor alliances (labor vs. capital). This matches empirical lobbying patterns in the US, EU.

Mathematical Derivation

Zero-Profit Condition (Revenue = Factor Payments)

By Euler's Theorem for CRS production functions:

$$Q_X = F_{X,L}L_X + F_{X,K}K_X$$

Multiply by P_X and substitute $w = P_X \cdot F_{X,L}$ and $r_X = P_X \cdot F_{X,K}$

$$P_X Q_X = wL_X + r_X K_X$$

This is the **zero-profit condition**: total revenue equals total factor payments.

Comparative Statics (Hat Notation)

Take the **total differential** and Apply **product rule**:

$$P_X dQ_X + Q_X dP_X = L_X dw + w dL_X + K_X dr_X + r_X dK_X$$

From the production function, $dQ_X = F_{X,L} dL_X + F_{X,K} dK_X$.

Multiply by P_X :

$$P_X dQ_X = w dL_X + r_X dK_X$$

so, $Q_X dP_X = L_X dw + K_X dr_X$,

Divide by $P_X Q_X$:

$$\frac{dP_X}{P_X} = \frac{w L_X}{P_X Q_X} \frac{dw}{w} + \frac{r_X K_X}{P_X Q_X} \frac{dr_X}{r_X}$$

- $\hat{x} \equiv \frac{dx}{x}$ (proportional change),
- $\theta_{LX} \equiv \frac{wL_X}{P_X Q_X}$ (labor's share of revenue/cost),
- $\theta_{KX} \equiv \frac{r_X K_X}{P_X Q_X}$ (capital's share)

Since **CRS** $\Rightarrow \theta_{LX} + \theta_{KX} = 1$, we get the **standard Jones hat-algebra equation**:

$$\hat{P}_X = \theta_{LX} \hat{w} + (1 - \theta_{LX}) \hat{r} \rightarrow \hat{r}_X = \frac{1}{1 - \theta_{LX}} (\hat{P}_X - \theta_{LX} \hat{w})$$

since $1 - \theta_{LX} < 0$, we can get the $\hat{r}_X > \hat{P}_X$ and $\hat{P}_X > \hat{w} > \hat{P}_Y$

This inequality is the core of the **Stolper-Samuelson / Specific-Factors magnification effect**:

Condition	Economic Meaning
$\hat{P}_X > 0$	Output price rises (e.g., trade liberalization, tariff removal, demand shock)
$\hat{w} < \hat{P}_X$	Nominal wage rises, but less than the output price (real wage in terms of good X falls)
$\hat{r}_X > \hat{P}_X$	Return to capital rises more than the output price (real return in terms of all goods rises)

The price increase generates a "revenue windfall" $\Delta P_X \cdot Q_X$. Because labor's share is only θ_{LX} , most of this windfall must flow to capital. The factor $1/(1 - \theta_{LX}) > 1$ **magnifies** the gap between $\hat{P}_X$ and $\hat{w}$, pushing $\hat{r}_X$ above $\hat{P}_X$.

1. Profit Maximization & Labor Demand

$$w = P_X \cdot MPL_X(L_X) = P_X F_{X,L}(K_X, L_X)$$

$$w = P_Y \cdot MPL_Y(L_Y) = P_Y F_{Y,L}(K_Y, L_Y)$$

Labor market clearing: $L_X + L_Y = \bar{L}$.

2. Comparative Statics (Hat Notation)

Let θ_{Li} = labor's share of revenue in sector i: $\theta_{Li} = \frac{wL_i}{P_iQ_i}$. Totally differentiate the equilibrium conditions:

$$\begin{aligned}\hat{w} &= \theta_{LX}\hat{P}_X + (1 - \theta_{LX})\hat{r}_X \\ \hat{w} &= \theta_{LY}\hat{P}_Y + (1 - \theta_{LY})\hat{r}_Y\end{aligned},$$

Solving the system yields the **key inequality** (assuming $\hat{P}_X > \hat{P}_Y = 0$):

$$\hat{P}_X > \hat{w} > \hat{P}_Y$$

3. Returns to Specific Factors

From zero-profit conditions:

$$\begin{aligned}\hat{r}_X &= \frac{1}{1-\theta_{LX}} \left(\hat{P}_X - \theta_{LX} \hat{w} \right) > \hat{P}_X \\ \hat{r}_Y &= \frac{1}{1-\theta_{LY}} \left(\hat{P}_Y - \theta_{LY} \hat{w} \right) > 0\end{aligned},$$

4. Real Returns Summary

Factor	Nominal Change	Real Return in X	Real Return in Y	Ambiguity?
Specific to X (K_X)	$\hat{r}_X > \hat{P}_X$	↑	↑↑	✗ Unambiguous gain
Specific to Y (K_Y)	$\hat{r}_Y < 0$	↓↓	↓	✗ Unambiguous loss
Mobile Labor (L)	$\hat{P}_X > \hat{w} > \hat{P}_Y$	↓	↑	✓ Ambiguous overall

Core Result: The **specific factor** in the **expanding sector** gains more than the **price rise**; the **specific factor** in the **contracting sector** **loses absolutely**; labor's welfare depends on consumption shares.